

Problem 12

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Problem 1 Sketch a possible graph of a function that satisfies all of the given properties. (You do not need to find a formula for the function.)

$$\begin{array}{lll}
 \lim_{x \rightarrow -2^-} f(x) = \infty & f(-2) = -5 & f(1) = 2 \\
 \lim_{x \rightarrow \infty} f(x) = -\infty & \lim_{x \rightarrow -\infty} f(x) = 4 & \lim_{x \rightarrow 5} f(x) = \infty \\
 \lim_{x \rightarrow 3} f(x) = 3 & f(3) = 1 & \\
 \lim_{x \rightarrow -2^+} f(x) = -5 & f(4) \text{ is undefined} & \lim_{x \rightarrow 4} f(x) = 3
 \end{array}$$